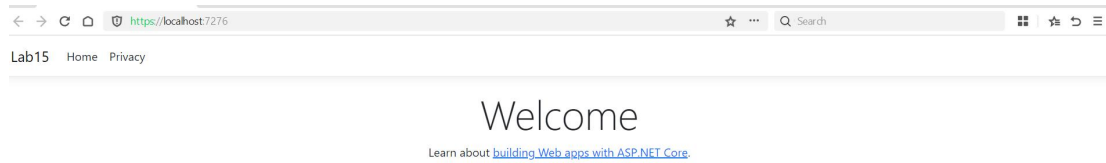


Muhamad Assem Ahmed

Lab 14

Part A



URL: <https://localhost:7276> Found the port in: Properties/launchSettings.json

Part B

B1 :

Build() doesn't just stop future registrations from taking effect — it actually locks the underlying ServiceCollection object and makes it throw the moment anything tries to add to it

```
Unhandled exception. System.InvalidOperationException: The service collection cannot be modified because it is read-only
    at Microsoft.Extensions.DependencyInjection.ServiceCollection.ThrowReadOnlyException()
    at Microsoft.Extensions.DependencyInjection.ServiceCollection.System.Collections.Generic.ICollection<Microsoft.Extensions.DependencyInjection.ServiceDescriptor>.Add(ServiceDescriptor item)
    at Microsoft.Extensions.DependencyInjection.ServiceCollectionServiceExtensions.AddTransient(IServiceCollection services, Type serviceType, Func`2 implementationFactory)
    at Microsoft.Extensions.Options.OptionsBuilder`1.Configure[TDep](Action`2 configureOptions)
    at Microsoft.Extensions.DependencyInjection.OptionsBuilderExtensions.ValidateOnStart[TOptions](OptionsBuilder`1 optionsBuilder)
    at Microsoft.Extensions.DependencyInjection.MetricsServiceExtensions.AddMetrics(IServiceCollection services)
    at Microsoft.Extensions.DependencyInjection.RoutingServiceCollectionExtensions.AddRoutingCore(IServiceCollection services)
    at Microsoft.Extensions.DependencyInjection.RoutingServiceCollectionExtensions.AddRouting(IServiceCollection services)
    at Microsoft.Extensions.DependencyInjection.MvcCoreServiceCollectionExtensions.AddMvcCore(IServiceCollection services)
    at Microsoft.Extensions.DependencyInjection.MvcServiceCollectionExtensions.AddControllersCore(IServiceCollection services)
    at Microsoft.Extensions.DependencyInjection.MvcServiceCollectionExtensions.AddControllersWithViewsCore(IServiceCollection services)
    at Microsoft.Extensions.DependencyInjection.MvcServiceCollectionExtensions.AddControllersWithViews(IServiceCollection services)
    at Lab15.Program.Main(String[] args) in D:\ITI .net\New folder\Lab15\Program.cs:line 13
```

B2 :

Nothing fails at compile time and nothing fails at startup either ASP.NET Core does not validate that every constructor requested type is registered until something actually tries to construct that controller so the failure should fail at the first time a request hits that controller

B3 :

Every request for a static file before it reaches later middleware. Since my middleware is registered AFTER UseStaticFiles() I predict it will not run at all for the CSS request 0 times only for the actual page request

Part C

Student Portal — Build #289

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Full Name	Year	GPA
Kareem Fouad	4	3.20
Nada Samir	1	3.60
Omar Hesham	3	2.80
Yara Adel	2	3.50

Part D

D2:

InvalidOperationException: Unable to resolve service for type 'Lab15.Models.StudentPortalContext' while attempting to activate 'Lab15.Controllers.HomeController'.

An unhandled exception occurred while processing the request.

InvalidOperationException: Unable to resolve service for type 'Lab15.Models.StudentPortalContext' while attempting to activate 'Lab15.Controllers.HomeController'.
Microsoft.Extensions.DependencyInjection.ActivatorUtilities.ThrowHelperUnableToResolveService(Type type, Type requiredBy)

Stack Query Cookies Headers Routing

InvalidOperationException: Unable to resolve service for type 'Lab15.Models.StudentPortalContext' while attempting to activate 'Lab15.Controllers.HomeController'.

```
Microsoft.AspNetCore.Diagnostics.DeveloperExceptionPageMiddleware[1]
An unhandled exception has occurred while executing the request.
System.InvalidOperationException: Unable to resolve service for type 'Lab15.Models.StudentPortalContext' while attempting to activate 'Lab15.Controllers.HomeController'.
   at Microsoft.Extensions.DependencyInjection.ActivatorUtilities.ThrowHelperUnableToResolveService(Type type, Type requiredBy)
```

D5:

The app started up with no error at all and served the Index page normally, returning the full, correct list of students from the database exactly as it did with the default (Scoped) registration
No exception, no startup refusal, no visible difference in behavior on this single request

D5:

that's bad news, not good news. A Singleton DbContext gets built once and then reused by every request forever, but DbContext is not thread safe and its change tracker accumulates state across requests

Part E

Lab15 Home Privacy

Owner: Muhamad Assem Ahmed | Stamp A: 43134d0f | Stamp B: 53b8be97

Welcome

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Full Name	Year	GPA
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Within a single load: Stamp A and Stamp B should NOT match, because the lifetime is Transient — every time the container is asked for IMohamedStampService, it constructs a brand-new instance, and the constructor generates a fresh GUID substring each time
Between loads: they should also differ from load to load, for the same reason

Part F

```
D:\ITI .net\New folder\Lab15\ x + -
Now listening on: http://localhost:5038
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: D:\ITI .net\New folder\Lab15
[START] /
[END] /
[START] /lib/bootstrap/dist/css/bootstrap.min.css
[START] /css/site.css
[START] /Lab15.styles.css
[START] /lib/jquery/dist/jquery.min.js
[START] /lib/bootstrap/dist/js/bootstrap.bundle.min.js
[START] /js/site.js
[END] /js/site.js
[END] /css/site.css
[END] /Lab15.styles.css
[END] /lib/jquery/dist/jquery.min.js
[END] /lib/bootstrap/dist/css/bootstrap.min.css
[END] /lib/bootstrap/dist/js/bootstrap.bundle.min.js
[START] /favicon.ico
[END] /favicon.ico
[START] /audit-27
[AUDIT] Muhamad Assem ahmed saw a request for /audit-27
[END] /audit-27
[START] /audit-27
[AUDIT] Muhamad Assem ahmed saw a request for /audit-27
[END] /audit-27
```

[START] lines appeared for the one home-page load: /, /css/site.css, /lib/bootstrap/dist/css/bootstrap.min.css, /Lab15.styles.css, /lib/jquery/dist/jquery.min.js, /lib/bootstrap/dist/js/bootstrap.bundle.min.js, /js/site.js. Loading one page produced more than one request because the browser first receives the HTML for /, then parses it and issues a separate HTTP request for every linked stylesheet and script referenced inside it. Extra observation: the [END] lines came back in a different order than the [START] lines (e.g. /css/site.css started third but ended first, while /lib/bootstrap/...css started second but ended second-to-last). That's because the browser fires the static-asset requests concurrently rather than one-at-a-time, so each one's await next() resumes whenever that particular request finishes

F.5 — Console output for the 404 audit path visit

```
D:\ITI .net\New folder\Lab15\ x + v
info: Microsoft.Hosting.Lifetime[0]
Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
Content root path: D:\ITI .net\New folder\Lab15
[START] /
[END] /
[START] /css/site.css
[START] /lib/jquery/dist/jquery.min.js
[START] /lib/bootstrap/dist/js/bootstrap.bundle.min.js
[START] /Lab15.styles.css
[START] /js/site.js
[START] /lib/bootstrap/dist/css/bootstrap.min.css
[END] /css/site.css
[END] /js/site.js
[END] /Lab15.styles.css
[END] /lib/bootstrap/dist/css/bootstrap.min.css
[END] /lib/jquery/dist/jquery.min.js
[END] /lib/bootstrap/dist/js/bootstrap.bundle.min.js
[START] /audit-27
[AUDIT] Muhamad Assem ahmed saw a request for /audit-27
[END] /audit-27
```

Paths that disappeared after moving the middleware after UseStaticFiles() Explanation: UseStaticFiles() now runs first and fully handles any request matching a physical file in wwwroot

Part G

G.1 : Deleting OnConfiguring removed the context's ability to decide its own connection details, but that's exactly what makes the app more flexible: the context no longer hardcodes an environment

G.2 : Failing at startup is better than failing on first request because a startup failure is loud, immediate, and impossible to miss you find out the moment you run the app, in a controlled setting

G.4 : If both the console app and the web app tried to own migrations, they'd each maintain their own separate migration history with no knowledge of the other's changes